

Eduardo Hafemann

<https://hafemanne.github.io/>

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Education

PhD in Mathematics

10/2023 – 10/2026

Universität Hamburg (UHH)

Supervisor: Prof. Dr. Melanie Graf.

Co-supervisor: Prof. Dr. Eleni-Alexandra Kontou (King's College London).

Thesis: Low-regularity metrics and Weakened Energy Conditions in Mathematical Relativity.

Msc Mathematics

08/2021 – 08/2023

Federal University of Santa Catarina - Florianópolis (UFSC)

Supervisor: Prof. Dr. Ivan Pontual Costa e Silva.

Dissertation: Geometry and Topology of Black Hole Horizons ([RI UFSC](#)).

Bsc Chemical Engineering

08/2016 – 08/2021

Federal University of Santa Catarina - Florianópolis (UFSC)

PAM Honors Program (2-year advanced track in mathematics)

Award: Highest GPA in Chemical Engineering, Graduating Class of August 2021.

Papers in Geometric Analysis and Mathematical Physics

Preprint

[1] M. Graf and E. Hafemann, “Coordinate isoperimetric mass and a positive mass theorem for distributional scalar curvature”, 2026, [arXiv:2610.08615 \[math.DG\]](#)

[2] E. Hafemann, “A low-regularity Riemannian positive mass theorem for non-spin manifolds with distributional curvature”, 2026, [arXiv:2602.03451 \[math.DG\]](#)

[3] E. Hafemann, E.-A. Kontou, and J. Procter, “Positive energy theorems and quantum energy inequalities”, 2026, (in preparation)

Published

[4] E. Hafemann and E.-A. Kontou, “Penrose inequality for integral energy conditions”, [Classical and Quantum Gravity](#) **42**, 195016 (2025)

[5] M. Calisti, M. Graf, E. Hafemann, M. Kunzinger, and R. Steinbauer, “Hawking’s singularity theorem for Lipschitz Lorentzian metrics”, [Commun. Math. Phys.](#) **406**, 207 (2025)

Papers in Applied Fields

Inverse Problems

- [6] A. De Cezaro, E. Hafemann, and A. Leitão, “A piecewise constant levelset approach for semi-blind deconvolution: application to barcode decoding”, 2026, [arXiv:2606.01156 \[math.NA\]](#)
- [7] A. De Cezaro, E. Hafemann, A. Leitão, and A. Osses, “A regularization method based on level-sets for the problem of crack detection from electrical measurements”, [Inverse Problems](#) **39**, 035009 (2023)
- [8] R. Filippozzi, E. Hafemann, J. C. Rabelo, F. Margotti, and A. Leitão, “A range-relaxed criteria for choosing the Lagrange multipliers in the Levenberg–Marquardt–Kaczmarz method for solving systems of non-linear ill-posed equations: Application to EIT-CEM with real data”, [Journal of Inverse and Ill-posed Problems](#) **31**, 267–292 (2023)
- [9] F. Margotti and E. Hafemann, “Range-relaxed strategy applied to the Levenberg–Marquardt method with uniformly convex penalization term in Banach spaces”, [Inverse Problems](#) **38**, 095001 (2022)

Particle Physics

- [10] B. C. Backes, E. Hafemann, I. Marzola, and D. P. Menezes, “Density-dependent quark mass model revisited: thermodynamic consistency, stability windows and stellar properties”, [Journal of Physics G: Nuclear and Particle Physics](#) **48**, 055104 (2021)

Chemical Engineering

- [11] T. Neitzel, C. S. Lima, E. Hafemann, D. A. A. Paixão, J. M. Junior, G. F. Persinoti, L. V. dos Santos, and J. L. Ienczak, “RNA-seq based transcriptomic analysis of the non-conventional yeast *Spathaspora passalidarum* during Melle-boinot cell recycle in xylose-glucose mixtures”, [Renewable Energy](#) **201**, 486–498 (2022)
- [12] E. Hafemann, R. Battisti, D. Bresolin, C. Marangoni, and R. A. F. Machado, “Enhancing chlorine-free purification routes of rice husk biomass waste to obtain cellulose nanocrystals”, [Waste and Biomass Valorization](#) **11**, 6595–6611 (2020)
- [13] E. Hafemann, R. Battisti, C. Marangoni, and R. A. Machado, “Valorization of royal palm tree agroindustrial waste by isolating cellulose nanocrystals”, [Carbohydrate Polymers](#) **218**, 188–198 (2019)
- [14] R. Battisti, E. Hafemann, C. A. Claumann, R. A. F. Machado, and C. Marangoni, “Synthesis and characterization of cellulose acetate from royal palm tree agroindustrial waste”, [Polymer Engineering and Science](#) **59**, 891–898 (2018)

Books

Inverse Problems

[15] F. Margotti, E. Hafemann, and L. M. Santana, *Implementação computacional da tomografia por impedância elétrica (Computational implementation of electrical impedance tomography)*, ISBN 978-85-244-0535-8 (Editora do IMPA, 2023)

Software Development

Developer of [FEIT](#), an open-source Python platform for solving forward and inverse problems in Electrical Impedance Tomography (EIT), developed for the Advanced Course at the 34th Brazilian Mathematics Colloquium (CBM).

Talks

Invited Talks

Geometry Seminar, Radboud University [1,2]	May 2026
Differential Geometry Seminar, Federal Univ. of Santa Catarina [2]	Mar 2026
Gravity Journal Club, King's College London [2]	Dec 2025
Mathematical Relativity Seminar, Universität Hamburg [4]	Apr 2025
Mathematical Relativity Seminar, Universität Hamburg [2]	Dec 2024

Contributed Talks

XII International Meeting on Lorentzian Geometry, IST Lisbon [1, 2]	July 2026
15th Central European Relativity Seminar, Radboud University [2]	Jan 2025
34th Brazilian Mathematics Colloquium, IMPA [9, 15]	Jul 2023

Research Stay & Training

Research Stay

King's College London, Prof. E.-A. Kontou, 1 week [3]	Dec 2025
King's College London, Prof. E.-A. Kontou, 1 week [4]	May 2024
Universität Wien, Profs. M. Kunzinger and R. Steinbauer, 1 week [5]	Dec 2023

Conference, Workshop, Masterclass and Meeting

Geometric Analysis: Parabolic and Elliptic Methods, KTH Stockholm	Jun 2026
Geometrical Aspects of Mathematical Relativity, Univ. of Copenhagen	Jun 2025
Introductory Workshop: New Frontiers in Curvature, SLMATH, Berkeley	Aug 2024
14th Central European Relativity Seminar, Universität Tübingen	Feb 2024

Teaching Experience

Theory of Distributions Teaching Assistant **10/2024 – 03/2025**
Responsible for creating exercise sheets and exercise classes.

Differential Geometry Teaching Assistant **08/2022 – 12/2022**
Assisted undergraduate mathematics students with homework and review sessions.

Language & Skills

Languages

Brazilian-Portuguese (Native), English (Fluent), German (A2).

Computer Skills

Python, FEniCS, R, MATLAB, Docker, Git, Django, HTML, CSS, JavaScript.